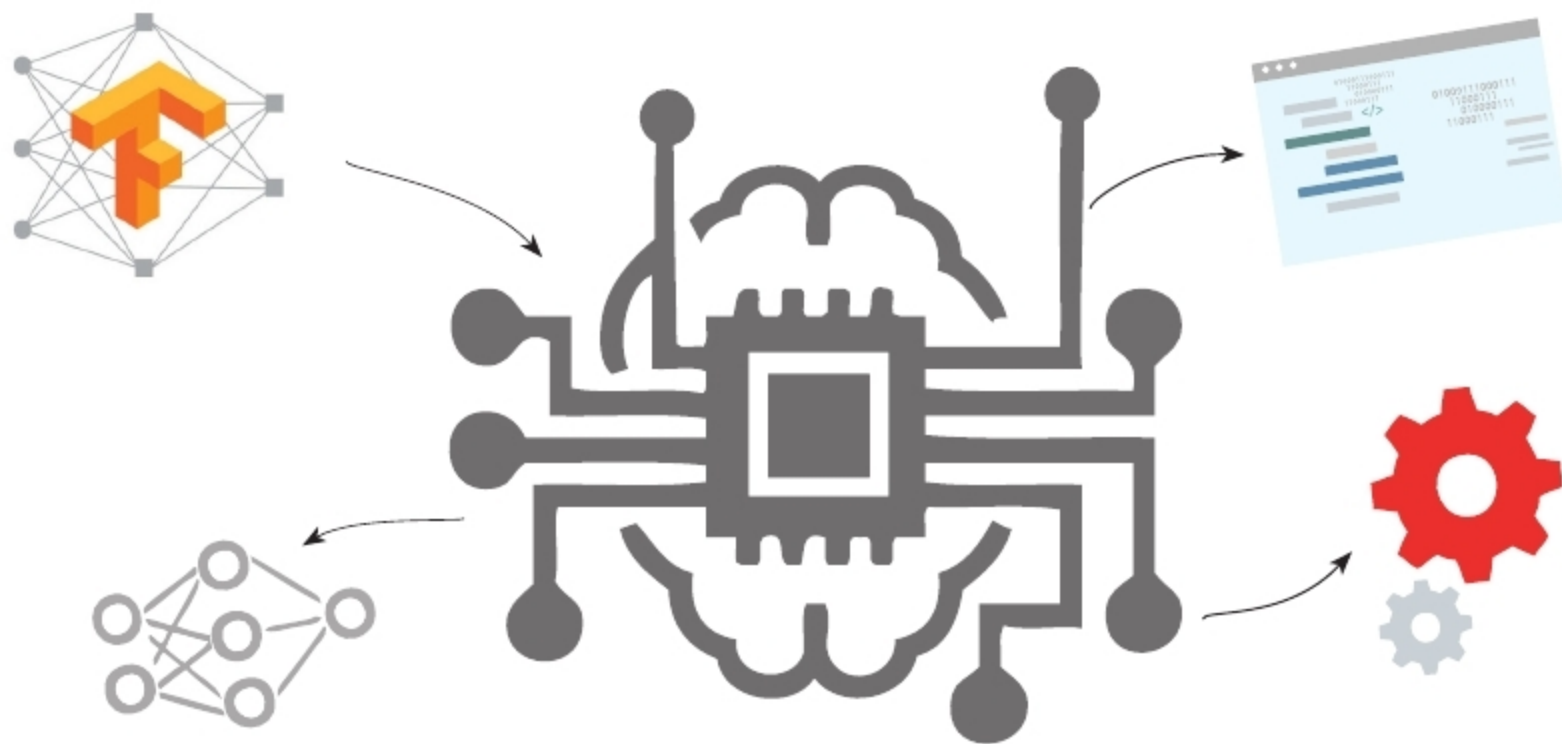


How TensorFlow Makes Machines Learn

This article covers TensorFlow, the machine learning tool developed by the Google Brain Team and open sourced in November 2015. Read on to learn what makes this tool suitable for machine learning applications and how to use it. You will understand why everyone at Google holds it in such high regard.



A few years ago, the Google Brain team developed a software library to perform machine learning across a range of tasks. The aim was to cater to the needs of the team's machine learning systems, those that were capable of building and training neural networks. The software was meant to help such systems detect and decipher patterns and correlations, just like the way human beings learn and reason.

In November 2015, Google released this library under the Apache 2.0 licence, making it open for use, providing everyone the opportunity to work on their own artificial intelligence (AI) based projects. By June 2016, 1500 repositories on GitHub mentioned the software, of which only five were from Google.

Working with TensorFlow

When you import TensorFlow into the Python environment, you get complete access over its classes, methods and symbols. You can take TensorFlow operations and arrange these into a graph of nodes called the computational graph. Typically, each node takes tensors as inputs and produces a corresponding output tensor. Values for the nodes get evaluated as and when a session is run. You can combine nodes with operations, which are also nodes in a certain form, to build more complicated computations.

Customise and improvise

To tune TensorFlow for your machine learning application, you need to construct the model such that it can take arbitrary inputs and deliver outputs accordingly. The way to do this with TensorFlow is to add variables, reflecting trainable parameters. Each of these has a type and an initial value, letting you tune your system to the required behaviour.

How do you know if your system is functioning exactly the way you intended it to? Simple... just introduce a loss function. TensorFlow provides optimisers that slowly change each variable so that loss functions can be minimised. There are also higher abstractions for common patterns, structures and functionality.

Multiple APIs for easier control

As a new user to any software, it is important to enjoy the experience. TensorFlow is built with that mindset, with the highest-level application program interface (API) tuned for easy learning and usage. With experience, you will learn how to handle the tool, and know which modification will result in changing the entire functionality and in what way this will happen.